VERA — Vision-Enabled Reasoning Agent — System Architecture

v1.70.1 — Calibrated fresh-frame visual servoing

Root-owned deployment controller (separate from VERA)
vera update + authenticated System GUI
/run/vera-updater/control.sock
peer-checked fixed operations
vera-updater.timer
vera-updater-check.service
local-time installation window
local Unix socket
vera-updater.service
/opt/vera-updater/src/updater
stage / deploy / verify / recover
manifest + verified assets
atomic journal
ZeroMQ IPC
verified full-duty kernel PWM
before thermal restart
GitHub stable releases
mazum01/desktop-assistant
HTTPS, exact tag/commit
/var/lib/vera-updater
protected backups + transaction state
post-reboot continuation
deployment.readiness RPC
loaded version/revision/PID
fresh telemetry + service/hardware health
NF-A6x25 PWM Fan
PWM ← GPIO13 (kernel HW PWM, 25 kHz)
TACH → GPIO6 (2 ppr)
tach
PWM
Hardware (Raspberry Pi 5)
Waveshare ESP32-C6-LCD-1.47
172×320 ST7789 SPI TFT
(BLE mouth/status display + BLE OTA)
reSpeaker XVF3800 output →
attached speaker
Pi Cam Module 3 Wide
(CSI-0)
reSpeaker XVF3800
(USB mic array)
Hailo AI Accelerator
(Hailo-8 / Hailo-10H, PCIe)
TMP117
(VERA3, I²C 0x48)
Pan servo + optional tilt
PCA9685 configurable channels
180deg direct-drive / legacy DS3218
configured safe travel limits
TMP102
(VERA, I²C 0x48)
SPI + GPIO
matched depth HEF
CSI
USB
PCIe
I²C
I²C/PWM
I²C
Drivers (src/)
vision/camera.py
(picamera2)
audio/input.py
perception/hailo_models.py
config/hailo/models.yaml
probe / select / download HEFs
python3 -m ...hailo_models --install
audio/version_announcer.py
perception/hailo_inference.py
(VStreams + InferModel API)
thermal/tmp117.py
motion/visual_servo.py
stable target lock + damped visual feedback
identified camera/servo Jacobian
config/tracking_calibration.json (local)
motion/servo_controller.py
USB
matched SCRFD HEF
TMP117 ID
perception/face_detector.py
(SCRFD decode + Haar fallback)
audio/tts.py
(espeak-ng)
perception/object_detector.py
(YOLOv11l on-chip NMS; YOLO26m+ONNX H8 fallback)
thermal/temperature_sensor.py
(probes TMP117 / TMP102)
thermal/fan_tach.py
(lgpio edge cb)
thermal/fan.py
(sysfs HW PWM
+ lgpio fallback)
renders to
audio/output.py
thermal/thermal_manager.py
matched ArcFace HEF
External processes
firmware/upload_esp32_display_ble.sh
+ nimbleota.py uploader
(BLE firmware push, no USB)
Web Browser
http://<pi>:8080
~/.local/share/desktop-assistant/
telemetry.db (SQLite ring)
faces.db (identity registry)
thumbs/ (64×64 JPEG face crops)
scripts/desktop-assistant
(CLI: ping / status / pan / say / watch / ...)
3rd-party subscribers
(dashboards, future ML pipelines)
HTTP/WS :8080
latest_frame()
writes telemetry.db
latest_chunk() + chunk_index()
ZMQ PUB→SUB
(cross-process)
ZMQ PUB→SUB
(cross-process)
writes faces.db
ZMQ REQ/SUB
latest_frame()
ZMQ SUB
BLE GATT (OTA)
VisionService
→ vision.frame_ready
← vision.capture_still
+ latest_frame()
latest_frame()
WebService (FastAPI)
GET / — dashboard SPA
GET /stream — MJPEG 640×360 feed
WS /ws — live JSON status
GET /api/faces — list (has_thumb flag)
GET /api/skills — registered skills
GET /api/faces/{id}/thumb — face JPEG
PUT /api/faces/{id} — rename
DEL /api/faces/{id} — delete one
DEL /api/faces — delete all
GET|PUT /api/settings/quiet-hours
GET|PUT /api/settings/audio
GET|PUT /api/settings/voice
GET|PUT /api/room
GET|PUT /api/music/* — proxied via IPCClient
GET|POST /api/podcasts/* — proxied via IPCClient
POST /api/say|pan|version|utterance
port 8080
IPCClient RPC
(music/podcast reads+actions)
IPCBridge (media)
PUB ipc:///tmp/desktop-assistant-media.pub
REP ipc:///tmp/desktop-assistant-media.rep
SUB ← core.pub (upstream)
RPCs: music.get_state/set_volume/...
podcast.search/subscribe/play/...
DisplayService
← system.startup_status / service.started / *.error
← display.spectrum / av.speaking_started / av.spoke
→ display.status / display.error
(BLE write to ESP32: status / mouth / eq)
PlaybackSpectrumService
← music.state_changed / podcast.playback
pw-record on sink monitor + FFT
→ display.spectrum
FaceService
← perception.faces / face.meet
→ av.say (greetings)
→ face.identified
cooldown-based re-greet
AudioCaptureService
→ audio.level / chunk
+ latest_chunk()
TelemetryService
← thermal.* / motion.position / audio.level
→ telemetry.flush
SQLite ring buffer
PerceptionService
← vision.frame_ready
→ perception.faces / perception.error
+ FaceEmbedder (ArcFace) + FaceRegistry
VoiceCommandService
← audio stream/VAD
→ voice.state / wake / transcript / intent
→ av.utterance (routed transcript)
wake-window + dialog context
systemd: desktop-assistant-thermal.service
IPCBridge (thermal)
PUB ipc:///tmp/desktop-assistant-thermal.pub
REP ipc:///tmp/desktop-assistant-thermal.rep
PodcastService
Apple Podcasts search + RSS subscriptions
← podcast.play/pause/resume/stop/refresh
→ podcast.playback / subscriptions_updated
MusicService
← music.play / stop / pause / next
← music.thumbs_up / thumbs_down / set_volume
Pandora / audio playback
ClockService
+ ClockAnnouncer
→ av.say (hourly/half-hourly)
respects QuietHours
NotificationService
← thermal.temp / perception.faces
← av.speaking_started / av.spoke
→ av.say (thermal alerts, absence alerts)
rate-limited · quiet-hours aware
SkillsService
← av.utterance / thermal.temp
→ av.say / vision.describe / av.tell_joke
→ face.meet / motion.pan_to / music.*
→ object.set_enabled / music.set_volume
→ tracking.set_face_tracking
SkillRegistry — first-match dispatch
17 skills: help · system_status · volume
quiet_hours · weather · reminder
smart_home · news + 9 others
ConfigField framework: enabled / config_schema
IPCBridge (core)
PUB ipc:///tmp/desktop-assistant.pub
REP ipc:///tmp/desktop-assistant.rep
SUB ← thermal.pub (upstream)
cmds: ping / topics / last / publish / status
AVService
← av.say / beep / utterance / announce_version
→ av.spoke / av.speaking_started / version_announced
ThermalService
— thermal.temp
— thermal.fan
— thermal.rpm
— thermal.critical
— thermal.error
TrackingService
← perception.faces (atomic frame time/geometry)
← motion.position / settings / limits_changed
→ motion.pan_to / tilt_to (optional)
← tracking.start_autotune / cancel_autotune
→ motion.calibration_lock / av.say
← av.spoke (announce before probes)
→ tracking.debug / autotune_progress / autotune_done
fresh observations only, settle then correct
Tilt: travel limits and center +/-20deg
idle random pan · speaking nod
MotionService
shared PCA9685, independent pan / tilt
50Hz bounded quintic profiles
← motion.pan_to / tilt_to / center / relax / stop
← motion.calibration_lock (expiring lease)
← motion.set_channels / set_limits / set_tilt_limits
→ motion.position / moved / settings / error
→ motion.channels_changed / limits_changed / tilt_limits_changed
RPC motion.get_status / motion.set_channels
ObjectService
← vision.frame_ready / object.set_enabled
→ object.detections
2fps · max_objects cap · BGR→RGB fix
latest_frame()
vision.mono_depth_map
latest_frame()
BLE GATT
status / mouth / eq
config R/W
suppresses
greetings
suppresses
announcements
suppresses
alerts
re-emit
extends
systemd: desktop-assistant-core.service
MonoDepthService
scdepthv3 on Hailo-8 / Hailo-10H
→ vision.mono_depth_map
relative 320×256 depth, 3 Hz
Arduino firmware
firmware/vera_display
(ESP32-C6 process boundary)
NimBLEOta service (BLE OTA, 0x8018)
RoomService
← room.set
→ room.updated / av.say
persists room name · visual scene
baseline histogram · sustained
divergence detection → TTS prompt
Core (src/core/)
version.py
(reads /VERSION)
media MessageBus
(in-process pub/sub)
QuietHours
(enabled / start / end
loaded from config/quiet_hours.yaml)
Service base
(start/tick/stop)
core MessageBus
(in-process pub/sub)
thermal MessageBus
(in-process pub/sub)